
Webbased-visualizer install guide

Version: 1.0

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Simona Filipović

Koen van Hove

Thomas Vrielink

Pieter Staal

Egbert Dijkstra

Inhoud

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1. Introduction

This document serves as the install guide for the webbased-visualizer and how you should connect it to the DEMKit simulator.

1.1 Changelog

Date	Version	Remarks
07-11-2019	1.0	Initial guide

2. Resources and literature

2.1 contact

Since the webbased-visualizer was a design project the creators cant be directly contacted. For general questions and support however you can mail demgroup-eemcs@utwente.nl. If the questions are too deep they know how to contact the creators of the visualizer.

In case of no direct response you can contact:

Gerwin Hoogsteen

Email: g.hoogsteen@utwente.nl

2.2 Code

We use two repositories for this project, one with the webbased-visualizer and one with our configuration files of DEMKit.

Specific resources:

DEMKit-config: https://github.com/PietPtr/DEMKit_SixConfig

Webbased-Visualizer: <https://gitlab.com/eggied97/webbased-visualizer>

2.3 Literature

For the literature we would advise you to read our design report (which is also provided with this zip file. In here we lay out our design choices and explain how the system is setup.

2.4 Third party documentation

The webbased-visualizer uses a couple of different tools and software. The following links should point to the documentation of those.

- NodeJS <https://nodejs.org/docs/latest-v11.x/api/>
- NPM <https://docs.npmjs.com/>
- Socket.IO <https://socket.io/docs/>

For project management we used a combination of github and gitlab. The documentation for those is available on the following links:

- Github <https://github.com/>
- Gitlab <https://about.gitlab.com/resources/>

3. Requirements

The Webbased-visualizer makes use of third-party libraries to make development easier but also to add more functionality. Instructions to run the visualizer follow in the next chapters.

3.1 Software

The following software is needed to run the webbased-visualizer:

- NodeJS
 - Download: <https://nodejs.org/en/download/current/>
 - Version: Latest
 - Notes: NPM is included.
- Docker
 - Download: <https://docs.docker.com/install/>
 - Version: latest
 - Notes: Optional, not needed for bare metal installation.

In addition, for development and code demonstrations we would advise to use the following software:

- Visual code
 - Download: <https://code.visualstudio.com/#alt-downloads>
 - Version: latest
 - Notes: It is an easy to use editor with integration with the terminal for quick and easy starting and stopping of the system. It also has an elaborate plugin market.

Besides those two code software packages we also use a couple of packages. Those can be installed through the command

`NPM install <package-name>`

Or alternatively, since we have a *package.json* we can also call *`NPM install`* and that should do the trick. The packages that we used were:

- Body-parser
 - Version: ^1.19.0
- Cors
 - Version: ^2.8.5
- Express
 - Version: ^4.17.1
- Request
 - Version: ^2.88.0
- Socket.io
 - Version: ^2.2.0

4. Running Webbased-visualizer

Since we have everything installed now we can commence to the next step, running the project. This can be done through couple of different ways, but in both cases we need DEMKit up and running. For the installation and configuration (where we also need the ALPG add-on) we guide you to the install guide of DEMKit, that is also provided with this ZIP file. So before we start with the webbased-visualizer we run demkit with the following command:

Python3 demkit.py -f ./ DEMKit Config -m demostreet network

For any errors we also guide you towards the DEMKit install guide.

For the webbased-visualizer we have two approaches. Running it bare metal or through a container software Docker.

4.1 bare metal

We can run the webbased-visualizer directly from node. The following command should then be executed:

Node index.js

This runs the webbased-visualizer on port 3002 and listens to DEMKit on *localhost:3001*, also the default values. If you navigate to *localhost:3002* you should see and interact with the webbased-visualizer.

4.2 Docker

To run it with docker takes some more steps, but is easier afterwards. First we need to edit our environment file. An example of this can be found in the *root* as *example.env*. You should rename this from *example.env* to *.env* and edit the values according to your install (this means that DEMKit can also be on another machine). After this there are again two steps, calling the Docker commands directly or going through the package manager.

4.2.1 Docker commands

In order to run a docker container we first need to build an image. This can be achieved through the following command:

Docker build -t webbased-visualizer .

We now have a docker image of our project called *webbased-visualizer*. You can change this name by changing the 'tag' (after *-t*). Now that we have an image we can run this through the following command:

Docker run --env-file=.env -p 3002:3002 -d -t webbased-visualizer

This command has a couple of parameters;

- `-env-file`
 - This is our environment file with all our configuration needed to connect it to DEMKit. We have configured this earlier.
- `-p`
 - This passes on the internal networkingport of docker to our machine. Since the webbased-visualizer runs on port 3002 we need to pass this on to our machine (3002:3002). If we would like to run the project on port 80 (website port) for example we need to change this to 80:3002
- `-d`
 - This enables us to run the container as a daemon.
- `-t webbased-visualizer`
 - This tells docker that we would like to run the just build image. If you choose to give it a different name you also need to change this.

4.2.2 NPM

If we would like to run the docker container through NPM we can easily do that trough the following command:

Npm start:deamon

This builds a docker image and runs it in daemon mode according to the instructions that are mentioned above.